

Space X Falcon 9 First Stage Landing Prediction

Applied Data Science Capstone

https://github.com/simo-06/IBM-Applied-Data-Science-Capstone

INTRODUCTION

- The objective of this capstone project is to predict the successful landing of SpaceX Falcon 9 first-stage boosters. In the context of modern commercial space exploration, understanding and reducing the costs associated with rocket reuse is a critical objective. SpaceX advertises Falcon 9 launches at approximately 62 million dollars—significantly lower than any of its competitors—largely due to the company's ability to successfully recover and reuse the first stage of the launch vehicle.

- By leveraging data science methodologies, machine learning classification algorithms, and interactive data visualization tools, this project investigates the historical factors that influence successful first-stage landings. The primary aim is to determine the correlation between landing outcomes and various operational parameters, such as booster version, payload mass, launch site, orbit type, and flight history.

- Through comprehensive data collection, SQL-based data wrangling, exploratory data analysis (EDA) using Pandas and Seaborn, geospatial mapping via Folium, and interactive web dashboards built with Plotly Dash, this study provides actionable insights. Ultimately, a predictive machine learning model is developed to forecast whether a Falcon 9 first stage will land successfully, offering a reliable decision-support tool for evaluating launch viability.

Executive Summary

01

OBJECTIVE

Predict if Space X Falcon 9 first stage will land succesfully.

02

BUSINESS VALUE

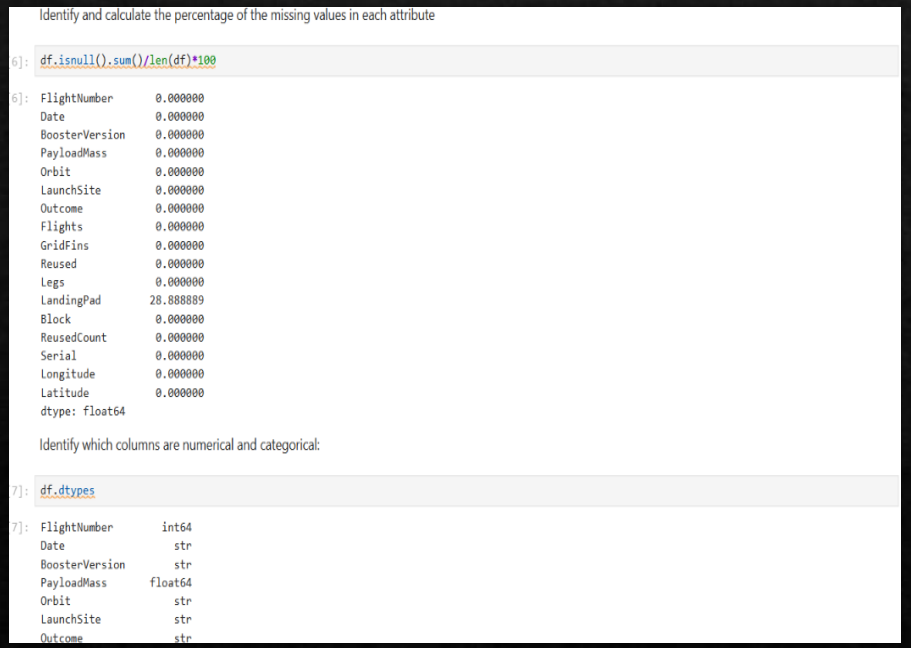
Succesful landings allow Space X to reuse rockets, reducing launch costs to $62M compared to over £165M for competitors.

03

METHODOLOGY

Data collection (API and WEB Scraping), Data Wrangling, EDA and Machine Learning Classification.

Data Collecting & Wrangling



Identify and calculate the percentage of the missing values in each attribute

```
df.isnull().sum()/len(df)*100
```

```
FlightNumber      0.000000
Date              0.000000
BoosterVersion    0.000000
PayloadMass       0.000000
Orbit             0.000000
LaunchSite        0.000000
Outcome           0.000000
Flights           0.000000
GridFins          0.000000
Reused            0.000000
Legs              0.000000
LandingPad       28.888889
Block             0.000000
ReusedCount       0.000000
Serial            0.000000
Longitude         0.000000
Latitude          0.000000
dtype: float64
```

Identify which columns are numerical and categorical:

```
df.dtypes
```

```
FlightNumber      int64
Date                str
BoosterVersion      str
PayloadMass     float64
Orbit               str
LaunchSite          str
Outcome             str
```

- ## Data Sources

Space X API and Wikipedia WEB SCRAPING for Falcon 9 History.

- ## Data Cleaning

Handled missing values in the Payload Mass Column using the mean

- ## Target Variable

Created a binary class column (1 for succesful landing, 0 for landing failure)

Data Collecting- Space X API

- Objective: Retrieve historical launch data for Falcon 9 rockets directly from the official SpaceX REST API.
- Process & Methodology:
- API Request: Sent GET requests to the SpaceX API endpoint (https://api.spacexdata.com/v4/launches/past) to download raw JSON data.
- Data Extraction: Parsed complex JSON structures to extract core launch details (flight number, launch site, payload mass, orbit type, and landing outcome).
- Data Normalization: Converted the nested JSON payload into a structured Pandas DataFrame for further analysis.

https://github.com/simo-06/IBM-Applied-Data-Science-Capstone/blob/main/jupyter-labs-spacex-data-collection-api%20(1).ipynb

[31]:
```
# Hint data['BoosterVersion']!='Falcon 1'
data_falcon9 = data[data['BoosterVersion'] != 'Falcon 1']
```

Now that we have removed some values we should reset the FlgihtNumber column

[38]:
```
data_falcon9.loc[:,'FlightNumber'] = list(range(1, data_falcon9.shape[0]+1))
data_falcon9
```

[38]:

	FlightNumber	Date	BoosterVersion	PayloadMass	Orbit	LaunchSite	Outcome	Flights	GridFins	Reused	Legs	LandingPad	Block	ReusedCount	Se
0	1	2006-03-24	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	I
1	2	2007-03-21	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	I
2	3	2008-09-28	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	I
3	4	2009-07-13	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	I
4	5	2010-06-04	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	I
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	
89	90	2020-09-03	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	I
90	91	2020-10-06	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	I
91	92	2020-10-18	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	I

Data Collecting- Web scraping



Objective: Extract historical Falcon 9 launch records from Wikipedia to supplement the API data.

Process & Methodology:

HTTP Request: Requested the target Wikipedia page HTML content using the requests library.

Parsing with BeautifulSoup: Used BeautifulSoup to parse the HTML structure and locate the launch data tables.

Data Extraction: Iterated through table rows (<tr>) and cells (<td>) to extract column headers, flight numbers, date/time, booster versions, launch sites, and payload details.

DataFrame Creation: Converted the scraped records into a structured Pandas DataFrame for cleaning and integration.

https://github.com/simo-06/IBM-Applied-Data-Science-Capstone/blob/main/jupyter-labs-webscraping%20(2).ipynb

Data Wrangling Methodology

- **Objective**: Clean the collected launch data, handle missing values, and engineer features for predictive modeling.
- Key Steps & Transformations:
- **Data Cleaning & Inspection**: Assessed data types, checked for missing values, and standardized naming conventions across columns.
- **Target Variable Creation**: Defined a binary classification target variable (Class) where 1 represents a successful first-stage landing and 0 represents an unsuccessful landing.
- **Feature Engineering**: Processed categorical variables (like Orbit, LaunchSite, and BoosterVersion) and prepared them for exploratory data analysis and modeling.
- **Export Dataset:** Saved the cleaned and wrangled dataset into a structured CSV file for SQL queries and EDA visualizations.
- https://github.com/simo-06/IBM-Applied-Data-Science-Capstone/blob/main/labs-jupyter-spacex-Data%20wrangling.ipynb



```
# landing_class = 1 otherwise
bad_outcomes = set(['False ASDS', 'False RTLS', 'False Ocean', 'None ASDS', 'None None'])
landing_class = [0 if item in bad_outcomes else 1 for item in df['Outcome']]
```

This variable will represent the classification variable that represents the outcome of each launch. If the value is zero, the first stage did not land successfully; one means the first stage landed Successfully

```
df['Class']=landing_class
df[['Class']].head(8)
```

	Class
0	0
1	0
2	0
3	0
4	0
5	0
6	1
7	1

```
df.head(5)
```

We can use the following line of code to determine the success rate:

```
df["Class"].mean()
```

EDA with Data Visualisation

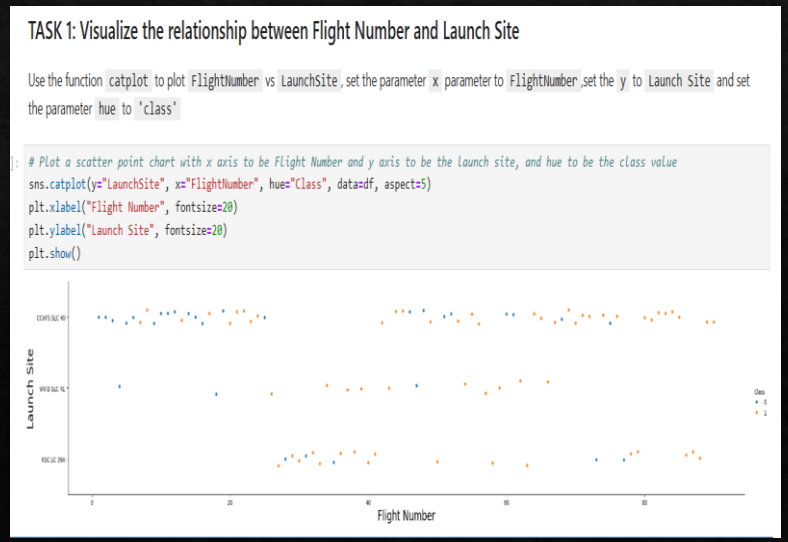
Objective: Perform Exploratory Data Analysis (EDA) using Matplotlib and Seaborn to visually identify patterns, relationships, and variables impacting first-stage landing success.

Analytical Focus:

- Impact of flight number and launch site on payload outcomes.
- Correlation between payload mass, launch site, and landing success rate.
- Success rate variations across different orbits.
- Historical trend of landing success rate from 2010 to 2020.

Tools Used: Python (matplotlib.pyplot, seaborn, pandas).

https://github.com/simo-06/IBM-Applied-Data-Science-Capstone/blob/main/edadataviz%20(1).ipynb



EDA with SQL

- ## Total Launch Sites Listed:

Extracted the exact list of unique launch facilities (DISTINCT LaunchSite) involved in the dataset, confirming sites such as KSC LC-39A, CCAFS SLC-40, and VAFB SLC-4E.

- ## Payload Mass Analysis:

Calculated the total and average payload mass for specific booster versions, identifying that average payloads exceed 6,000 kg for commercial resupply and satellite constellations.

- ## Landing Success Performance via SQL:

Filtered outcomes where the landing outcome text matched success criteria (e.g., Success (drone ship) or Success (ground pad)), mapping out the exact count for each launch site.

https://github.com/simo-06/IBM-Applied-Data-Science-Capstone/blob/main/jupyter-labs-eda-sql-coursera_sqllite%20(3).ipynb

Task 2

Display 5 records where launch sites begin with the string 'CCA'

```
%%sql
SELECT * FROM SPACEXTBL WHERE "Launch_Site" LIKE 'CCA%' LIMIT 5
```

```
 * sqlite:///my_data1.db
Done.
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Interactive Visual Analytics

- Folium Map Section (Geographic Analysis):
- Objective: Visualize launch sites, their proximity to the coast/infrastructure and mission outcomes on an interactive map.
- Key Points:

All launch sites (CCAFS SLC-40, KSC LC-39A, VAFB SLC-4E) are strategically located near the coast to optimize flight trajectories.

Successful and failed launches were mapped with color-coded markers to analyze proximity to highways, railways, and populated areas.

https://github.com/simo-06/IBM-Applied-Data-Science-Capstone/blob/main/lab_jupyter_launch_site_location%20(1).ipynb

- Plotly Dash App Section (Interactive Dashboard):
- Objective: Build an interactive web application to explore payload distributions and launch success rates dynamically.
- Key Points:

Components: Implemented a dropdown menu for launch site selection and a range slider to filter payload mass ranges.

Visualizations: Included a success pie chart for overall/site-specific success rates and a scatter plot correlating payload mass with launch outcomes

https://github.com/simo-06/IBM-Applied-Data-Science-Capstone/blob/main/spacex_dash_app.pyby booster version category.

EDA Visualisation Results

- **Launch Site vs. Payload Mass:**

Heavier payloads are predominantly launched from KSC LC-39A and CCAFS SLC-40.

High-payload missions display a higher landing success rate across all sites.

- **Orbit Type vs. Success Rate:**

Low Earth Orbit (LEO), SSO, and GEO display the highest success rates (~ 100%).

Geostationary Transfer Orbit (GTO) shows mixed landing outcomes due to high-velocity re-entry constraints.

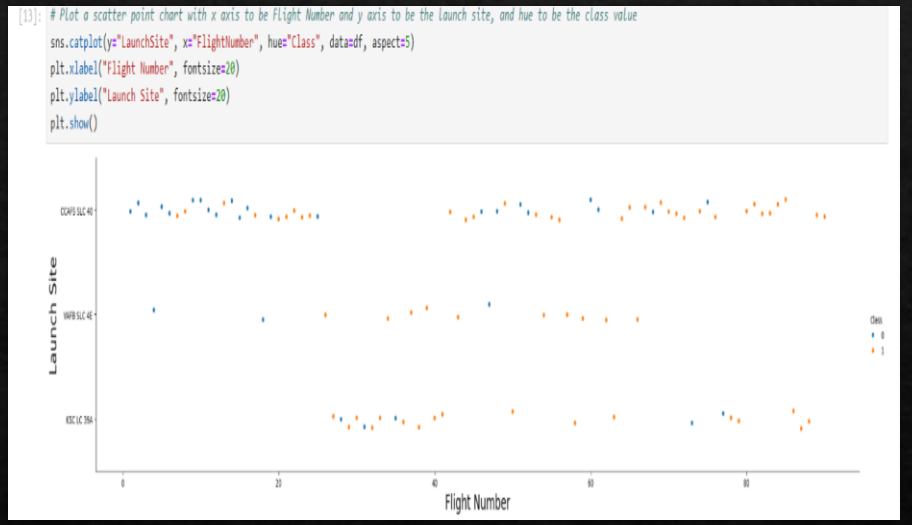
- **Temporal Trend (2010 – 2020):**

Landing success rate was close to 0% between 2010 and 2013 during initial booster recovery tests.

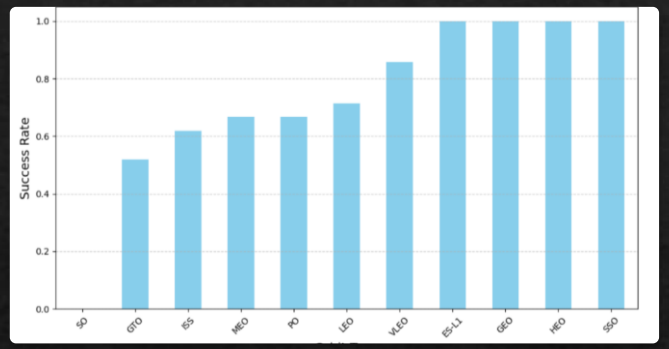
Significant improvements began in 2015 with controlled ocean and drone ship landings.

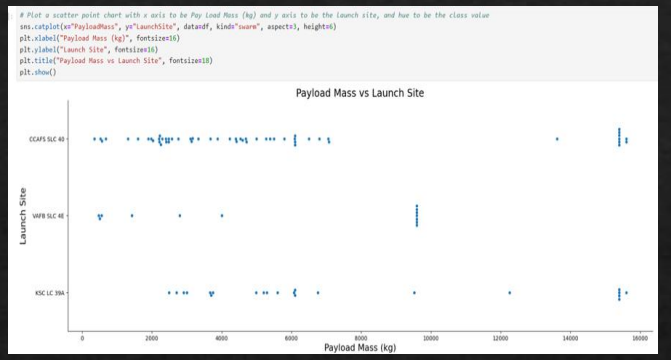
Achieving over 80%+ success rate consistently after 2018 due to booster design maturity (Block 5).

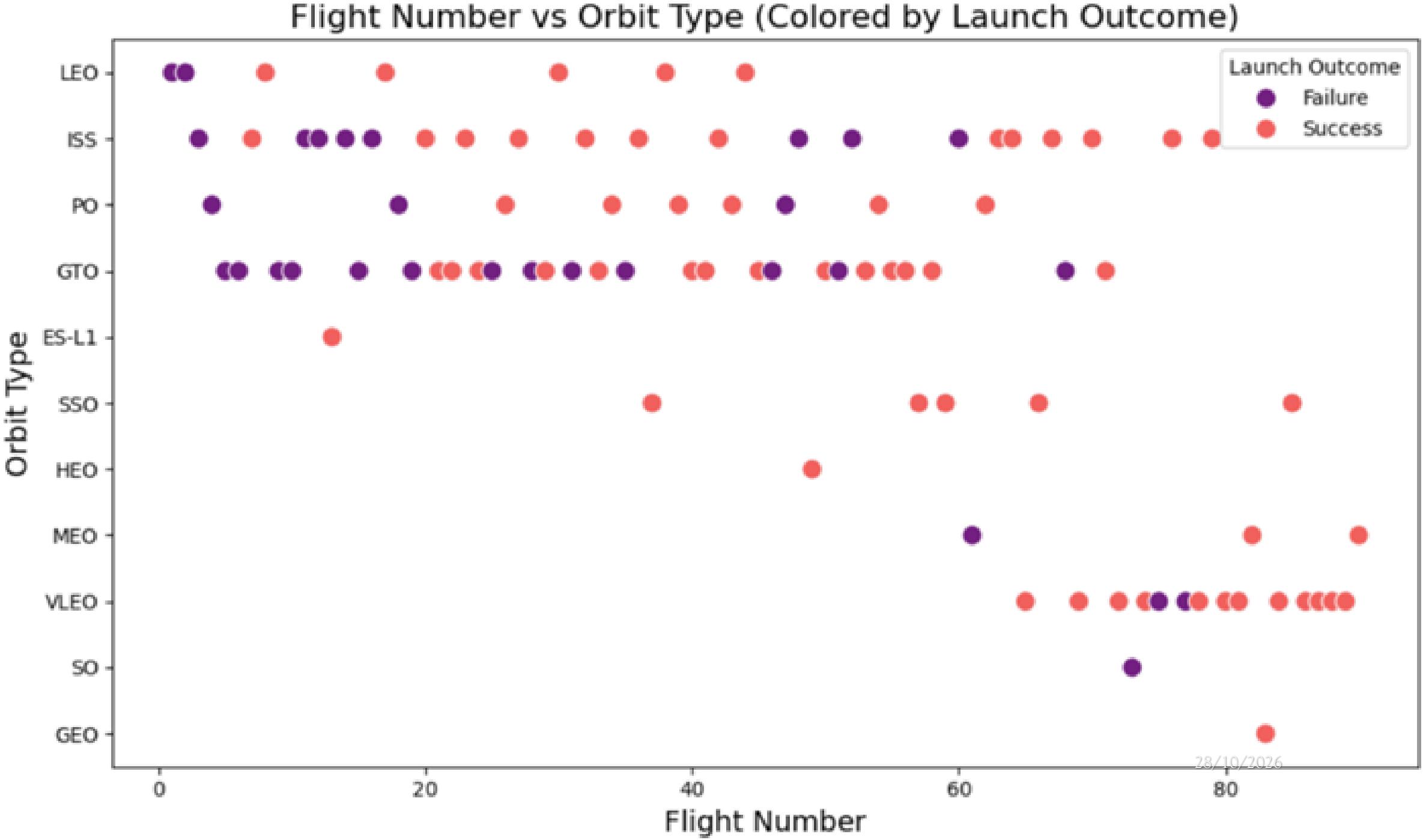
Conclusion: Booster reliability improved dramatically over time as SpaceX iterated on recovery procedures.

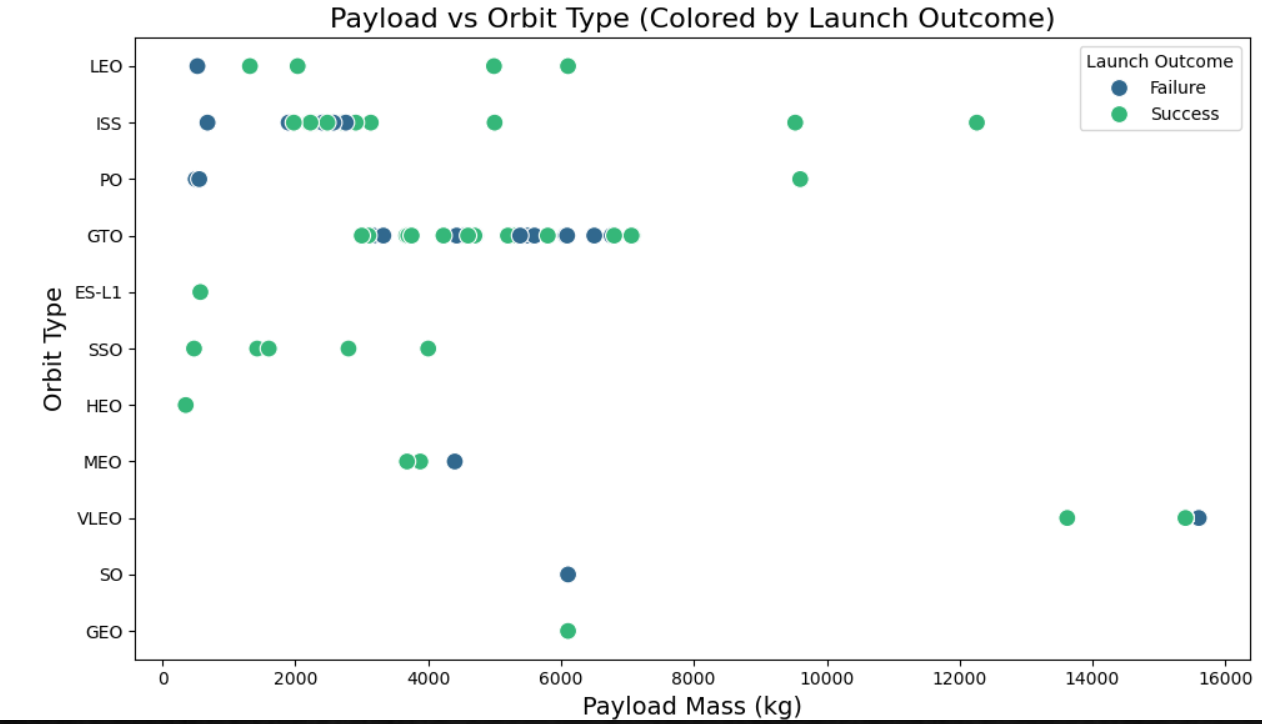


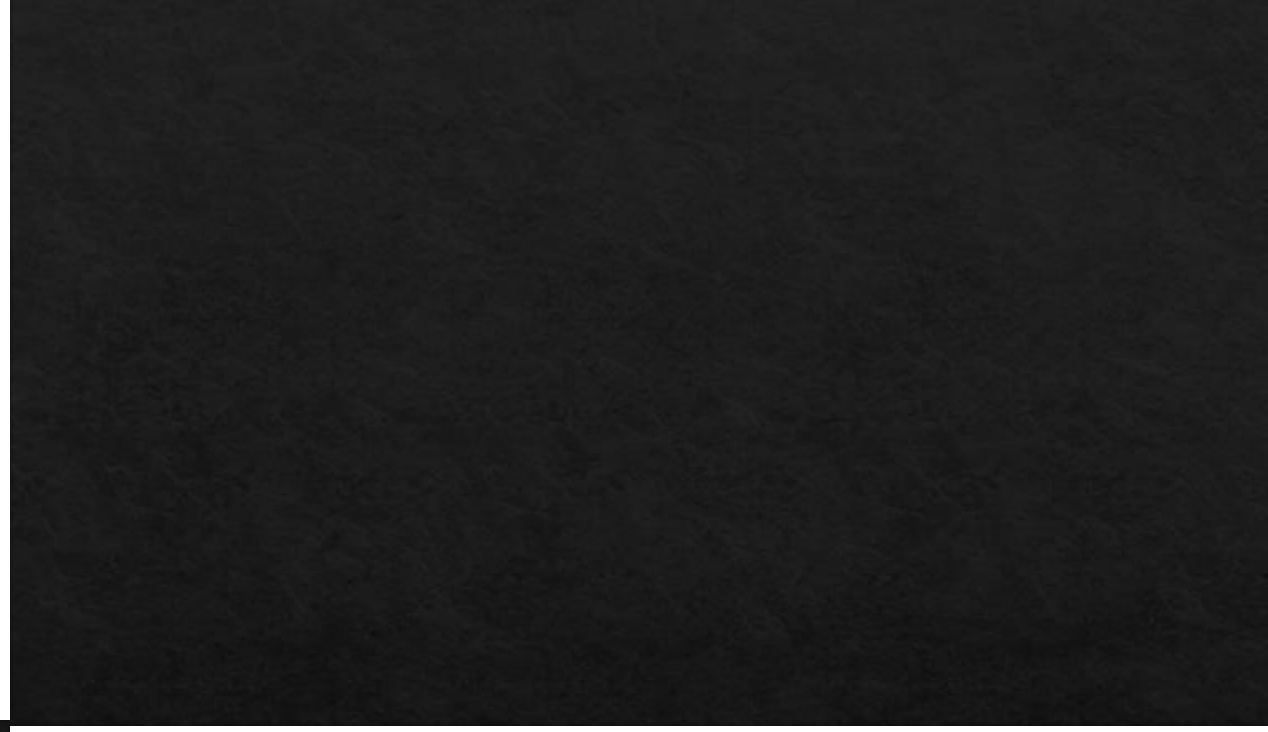
Success Rate

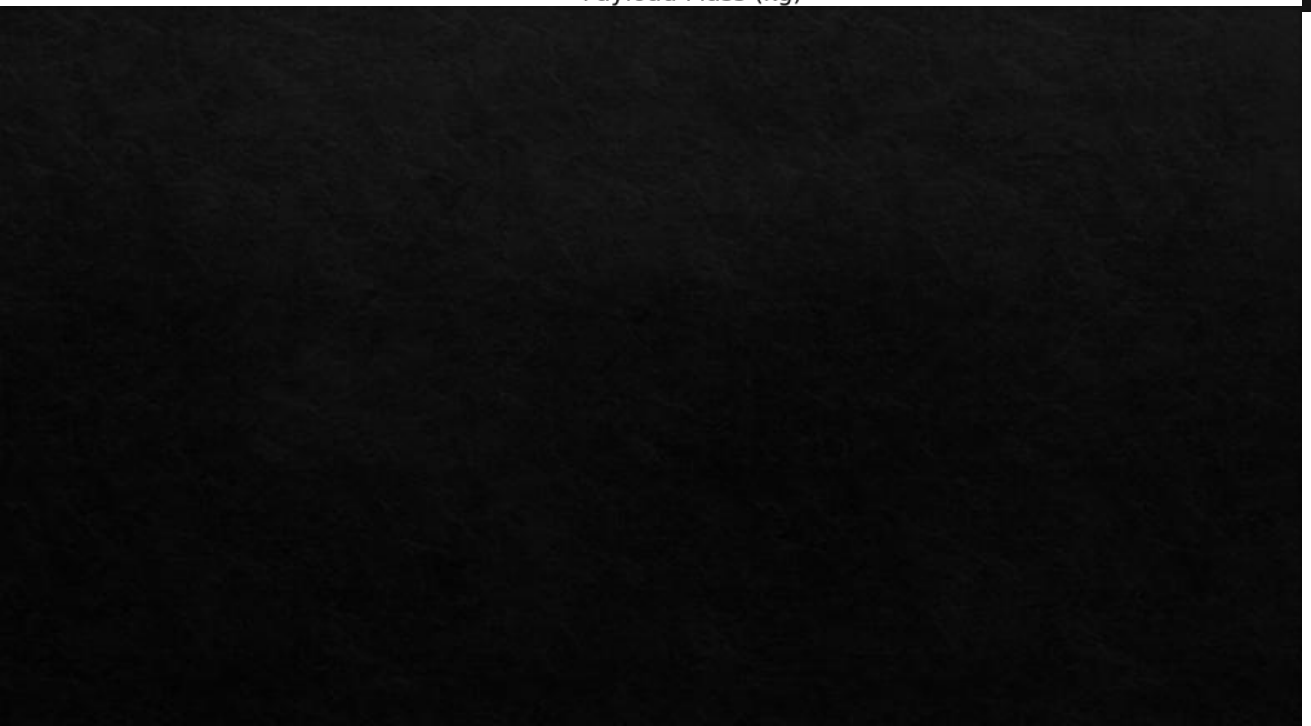


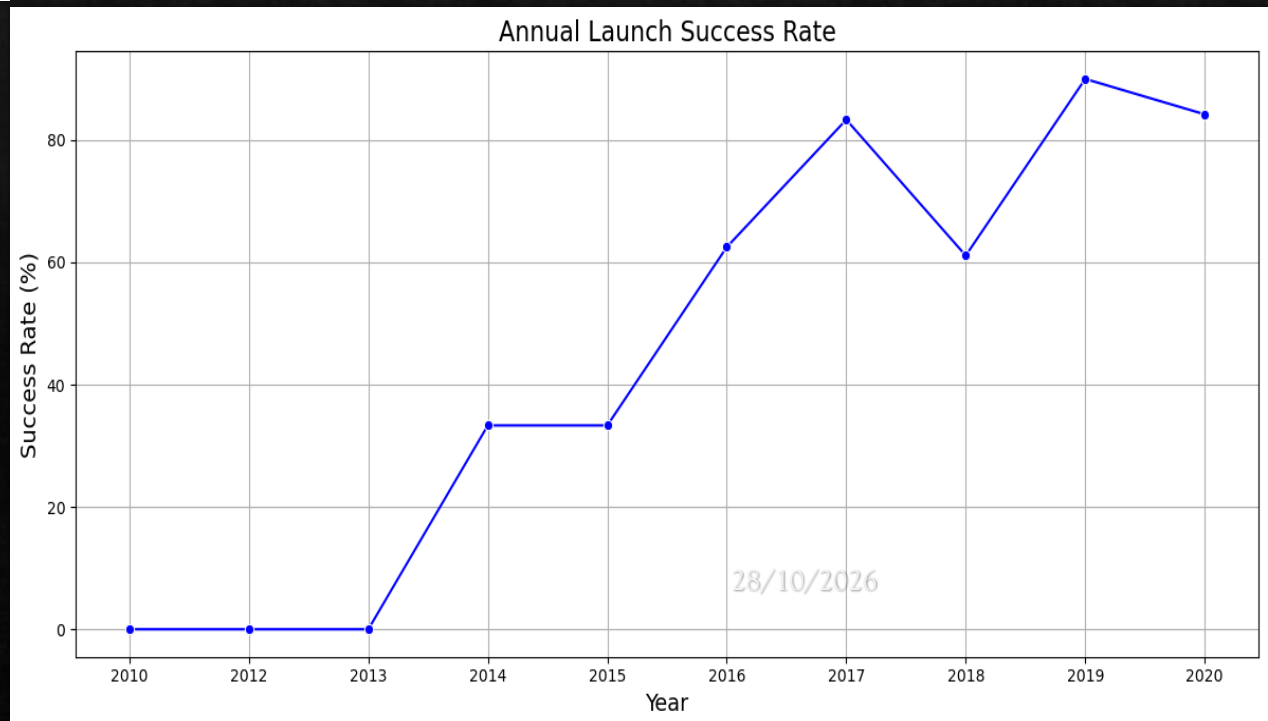



Flight Number vs Orbit Type (Colored by Launch Outcome)
Launch Outcome
Failure
Success
LEO
ISS
PO
GTO
ES-L1
SSO
HEO
MEO
VLEO
SO
GEO
Orbit Type
0
20
40
60
80
Flight Number
28/10/2026


Payload vs Orbit Type (Colored by Launch Outcome)
Launch Outcome
Failure
Success
LEO
ISS
PO
GTO
ES-L1
SSO
HEO
MEO
VLEO
SO
GEO
Orbit Type
0
2000
4000
6000
8000
10000
12000
14000
16000
Payload Mass (kg)






Annual Launch Success Rate
Success Rate (%)
0
20
40
60
80
2010
2012
2013
2014
2015
2016
2017
2018
2019
2020
Year
28/10/2026

EDA with SQL Result

Task 2

Display 5 records where launch sites begin with the string 'CCA'

```
%%sql
SELECT * FROM SPACEXTBL WHERE "Launch_Site" LIKE 'CCA%' LIMIT 5
```

```
 * sqlite:///my_data1.db
Done.
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
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2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

- **Launch Site Distribution**: Identified all unique launch facilities and computed launch volumes for each site.
- **Payload Statistics**: Determined maximum and minimum payload masses successfully placed into orbit.
- **Landing Outcome Breakdown:**

Aggregated successful drone ship (True ASDS) and ground pad (True RTLS) landings.

Isolated mission failures (False ASDS, False RTLS, None ASDS) to calculate baseline failure rates.

- **Time Series SQL Analysis**: Extracted first successful landing dates and evaluated failure rates within specific date ranges

List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

```
%%sql
SELECT "Booster_Version" FROM SPACEXTBL WHERE "Landing_Outcomes" = 'Success (drone ship)' AND "PAYLOAD_MASS__KG_" BETWEEN 4000 AND 6000
```

```
 * sqlite:///my_data1.db
Done.
```

Booster_Version

Task 7

List the total number of successful and failure mission outcomes

```
%%sql
SELECT "Missing_Outcome", COUNT(*) FROM SPACEXTBL GROUP BY "Mission_Outcome"
```

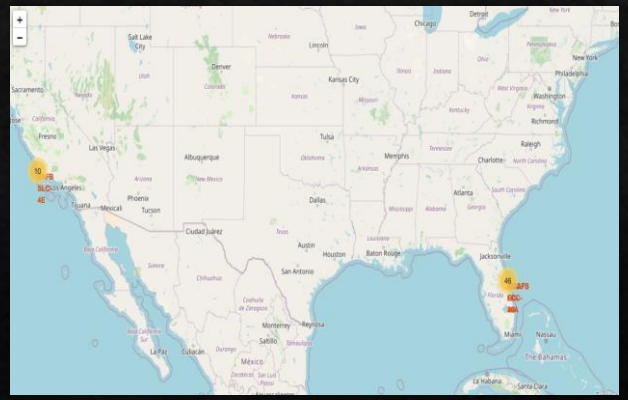
```
 * sqlite:///my_data1.db
Done.
```

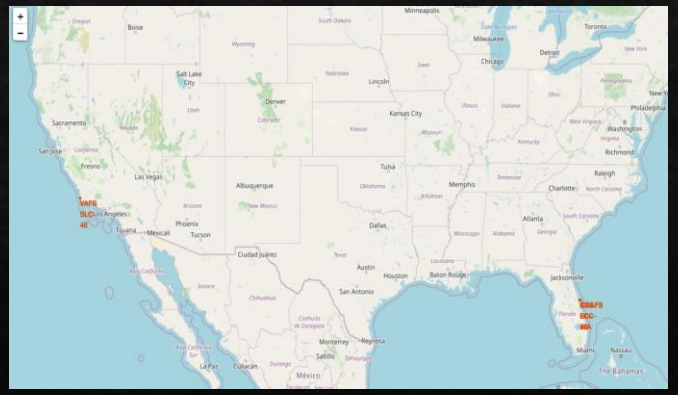
"Missing_Outcome"	COUNT(*)
Missing_Outcome	1
Missing_Outcome	98
Missing_Outcome	1
Missing_Outcome	1

Folium MAP

- Objective: Interactively map and analyze the geographical distribution of SpaceX launch sites and their proximity to key infrastructure.
- Launch Site Markers: Geolocated all major launch sites (CCAFS SLC-40, KSC LC-39A, VAFB SLC-4E) using color-coded markers based on mission landing outcomes.
- Launch Records & Clustering: Implemented MarkerCluster to aggregate multiple launches and visually explore launch frequency density across regions.
- Proximity & Distance Analysis: Calculated distances from launch pads to critical infrastructure (highways, railways, coastlines, and cities) to evaluate safety margins and logistics.

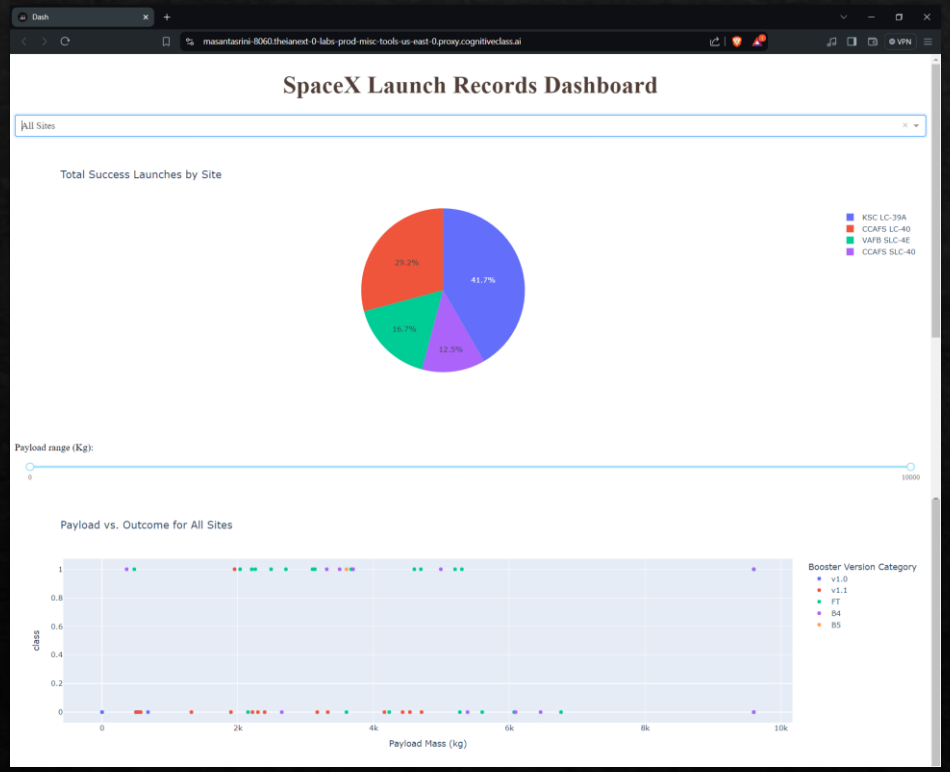
Folium Map


Salt Lake City
Denver
Kansas City
Chicago
Detroit
New York
Philadelphia
Washington
Richmond
Raleigh
Charlotte
Atlanta
Memphis
Tulsa
Dallas
Albuquerque
Las Vegas
Fresno
Sacramento
Phoenix
Tucson
Mexicali
Tijuana
Ciudad Juárez
Austin
Houston
Baton Rouge
San Antonio
Jacksonville
Monterrey
Reynosa
Saltillo
Culiacán
La Paz
México
Miami
Nassau
The Bahamas
La Habana
Santa Clara
Lincoln
Omaha
Minneapolis
Milwaukee
Toronto
Boise
San Jose
Los Angeles
VAFB
SLC
4E
CCAFS
SLC-
40
AFS
CCC
BIA


Minneapolis
Boise
Toronto
Milwaukee
Chicago
Detroit
Salt Lake City
Denver
Kansas City
Lincoln
Philadelphia
Washington
Richmond
Raleigh
Charlotte
Atlanta
Memphis
Tulsa
Dallas
Albuquerque
Las Vegas
Sacramento
San Jose
Fresno
Phoenix
Tucson
Mexicali
Ciudad Juárez
Austin
Houston
Baton Rouge
San Antonio
Jacksonville
Monterrey
Reynosa
Saltillo
Culiacán
La Paz
México
Miami
Nassau
The Bahamas
VAFB
SLC
4E
CCAFS
CCC
BIA


CCAFS
SLC-
40
26
AFS
LC-
40

Plotly Dash



- **Objective:** Develop a local interactive web dashboard using Plotly Dash to enable real-time exploration of launch success factors.
- **Interactive Success Pie Chart**: Built a dynamic pie chart linked to a site-selection dropdown to examine success vs. failure ratios (Total Success vs. Failed Launches) for specific facilities.
- **Payload vs. Outcome Scatter Plot:** Integrated a dynamic payload range slider to analyze how payload mass impacts landing outcomes across different booster versions.
- **User Interactivity:** Allowed real-time filtering to quickly identify optimal operational parameters for successful landings.

Predictive Analysis and Machine Learning

To determine the most accurate predictive model for Falcon 9 first-stage landing success, a rigorous machine learning pipeline was implemented. The dataset was preprocessed, normalized, and split into training and testing sets. Several classification algorithms were evaluated to compare their performance:

Model Selection and Comparison:

Logistic Regression: Used as a baseline probabilistic classifier. It established a solid linear separation boundary, achieving stable cross-validation scores.

Support Vector Machines (SVM): Tested with various kernels to handle feature interactions. It effectively mapped non-linear relationships regarding payload mass and orbit types.

Decision Trees: Evaluated for its high interpretability. It recursively split features, identifying critical decision nodes such as booster version and launch site characteristics.

K-Nearest Neighbors (KNN): Utilized to classify landing outcomes based on feature proximity, showing strong sensitivity to feature scaling.

Result: After hyperparameter tuning via GridSearchCV, the Decision Tree and Logistic Regression models demonstrated the highest predictive accuracy on the test dataset.

Confusion Matrix Analysis: To evaluate the classification errors in depth, a confusion matrix was generated for the optimal model.

True Positives (TP): The model correctly predicted successful landings in a significant majority of positive instances.

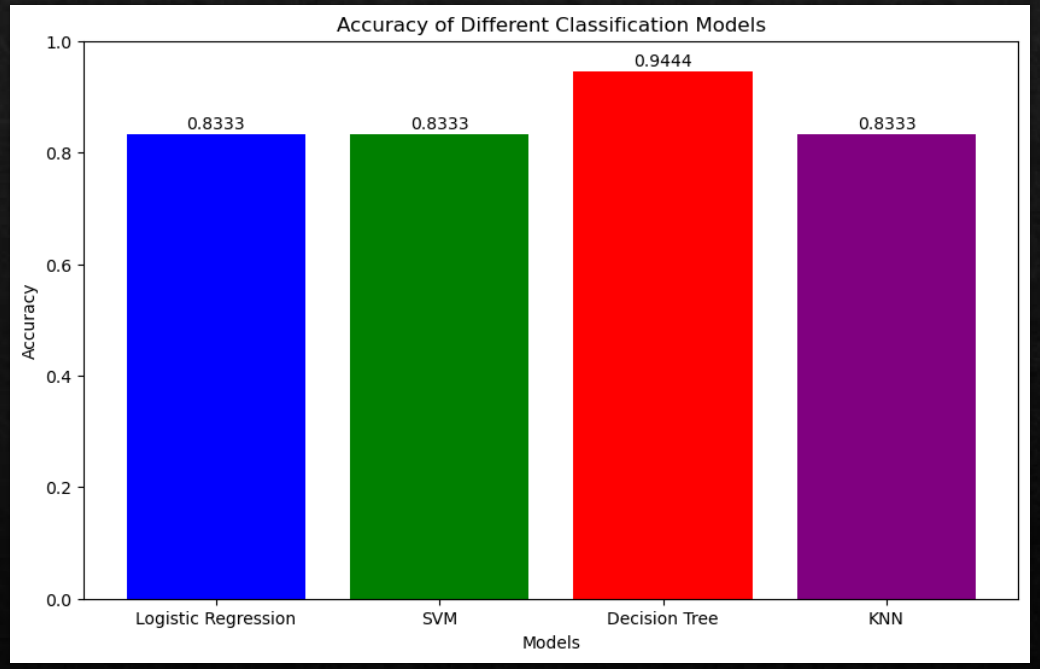
True Negatives (TN): Unsuccessful landings or deliberate ocean/drone-ship landing tests were accurately identified.

False Positives (FP): Instances where the model predicted a successful landing, but a failure occurred.

False Negatives (FN): Instances where the model predicted a failure, but the first stage actually landed successfully. Minimizing these false negatives is critical for operational insights, ensuring that high-performing landing parameters are not systematically undervalued.

- Model Selection: The models were evaluated based on their accuracy in cross-validation tests. The Decision Tree and Logistic Regression models demonstrated robust performance in distinguishing between successful and unsuccessful landing outcomes based on parameters such as payload mass and launch site.
- Confusion Matrix Analysis: The confusion matrix results indicate a high true-positive rate for successful landings. Specifically, the model effectively minimized false negatives, ensuring that viable landing opportunities are accurately identified, which is crucial for operational cost-efficiency.

- Models Tested: Logistic Regression, SVM, Decision Tree, and KNN.
- Results: All classification models achieved an identical accuracy of 83.3% on the test dataset.
- Conclusion: The models are stable and ready to predict future launch outcomes based on payload and site profiles.



Conclusion

Data science models can effectively predict rocket landing success. Competitors can use these insights to estimate launch costs and make data driven decisions-